

The empirical recurrence for OEIS A298708: recovering a conjecture that is not written down

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Abstract

OEIS A298708 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A298708 is “Number of $n \times 4$ 0..1 arrays with every element equal to 1, 3, 4, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 13, 2, 11, 32, 87, 110, 385, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jan 25 2018 08:34:03, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 72$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 144$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	2	c_{19}	2938619	c_{37}	21516579	c_{55}	−663286
c_2	4	c_{20}	8315408	c_{38}	68468532	c_{56}	−781002
c_3	41	c_{21}	3903977	c_{39}	36060897	c_{57}	−189951
c_4	−85	c_{22}	−8818573	c_{40}	−9585768	c_{58}	287809
c_5	−198	c_{23}	−24396609	c_{41}	−25800601	c_{59}	203426
c_6	−711	c_{24}	−10362544	c_{42}	−21439709	c_{60}	112313
c_7	1602	c_{25}	19668600	c_{43}	1301726	c_{61}	−26136
c_8	4069	c_{26}	53595809	c_{44}	−2748036	c_{62}	−41548
c_9	7308	c_{27}	21600261	c_{45}	7074151	c_{63}	−34480
c_{10}	−17893	c_{28}	−32291802	c_{46}	1810423	c_{64}	−29452
c_{11}	−48550	c_{29}	−88838466	c_{47}	10860190	c_{65}	−8208
c_{12}	−51950	c_{30}	−35014185	c_{48}	381867	c_{66}	−144
c_{13}	134746	c_{31}	38603465	c_{49}	−1530007	c_{67}	7232
c_{14}	381284	c_{32}	111197775	c_{50}	−7204019	c_{68}	6016
c_{15}	278013	c_{33}	44336645	c_{51}	−1160836	c_{69}	2816
c_{16}	−730868	c_{34}	−33551688	c_{52}	955979	c_{70}	1024
c_{17}	−2093682	c_{35}	−103749741	c_{53}	2737715		
c_{18}	−1167771	c_{36}	−44750783	c_{54}	391864		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 28 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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